Every Kind of Analysis You Could Possibly Want to Do

Alyssa Brock and Avneet Virk

Data Exploration and Assessing Assumptions

1. Data Structure and Initial Inspection

Understanding the dataset before conducting any analysis.

- Dataset dimensions (rows and columns)
- Variable types (numeric, categorical, ordinal, date/time)
- Continuous vs. discrete variables
- Identifying unique observations and duplicate records
- Identifying unique values and categories
- Data types and class conversion
- Data consistency and formatting
- Identifying invalid or impossible values
- Data validation and integrity checks

R functions: str(), glimpse(), summary(), head(), dim(), nrow(), ncol(), unique(), table()

2. Descriptive Statistics and Summary Measures

Summarizing the characteristics and distribution of variables.

- Mean, median, and mode
- Minimum and maximum
- Range
- Variance and standard deviation
- Interquartile range (IQR)
- Quartiles and percentiles
- Coefficient of variation
- Skewness
- Kurtosis
- Five-number summary
- Grouped summary statistics
- Weighted descriptive statistics

2.1 Measures of Central Tendency Measures used to identify the center or typical value of a dataset.

- `mean()` – Arithmetic mean
- `median()` – Median
- `Mode()` – Mode (from `DescTools`; base R does not have a built-in statistical mode function)
- `weighted.mean()` – Weighted mean
- `Hmisc::wtd.mean()` – Weighted mean with additional options
- `DescTools::Mode()` – Statistical mode

2.2 Measures of Variability and Dispersion Measures used to quantify the spread or variability of observations.

- `var()` – Variance
- `sd()` – Standard deviation
- `IQR()` – Interquartile range
- `range()` – Minimum and maximum
- `diff(range())` – Range width
- `mad()` – Median absolute deviation
- `DescTools::Var()` – Variance
- `DescTools::CoefVar()` – Coefficient of variation
- `matrixStats::rowVars()` – Row-wise variance
- `matrixStats::colVars()` – Column-wise variance

2.3 Measures of Position and Distribution Measures used to describe the relative position and distribution of observations.

- `min()` – Minimum
- `max()` – Maximum
- `quantile()` – Quantiles
- `rank()` – Observation ranks
- `scale()` – Standardization
- `ecdf()` – Empirical cumulative distribution function
- `summary()` – Summary statistics
- `fivenum()` – Five-number summary
- `Hmisc::describe()` – Detailed descriptive statistics
- `psych::describe()` – Comprehensive descriptive statistics

2.4 Shape of Distribution Measures used to describe the shape and characteristics of a distribution.

- `e1071::skewness()` – Skewness
- `moments::skewness()` – Skewness
- `e1071::kurtosis()` – Kurtosis

- `moments::kurtosis()` – Kurtosis
- `PerformanceAnalytics::skewness()` – Skewness
- `PerformanceAnalytics::kurtosis()` – Kurtosis

Note: Different packages can use different conventions for calculating skewness and kurtosis, so be mindful of the definitions.

2.5 Frequency and Categorical Summaries Functions used to summarize categorical variables and frequency distributions.

- `table()` – Frequency counts
- `prop.table()` – Proportions
- `margin.table()` – Marginal frequencies
- `addmargins()` – Add row and column totals
- `xtabs()` – Cross-tabulation
- `janitor::tabyl()` – Tidy frequency tables
- `janitor::adorn_totals()` – Add totals to frequency tables
- `dplyr::count()` – Count observations by group

2.6 Grouped and Advanced Summaries Functions used to calculate descriptive statistics across groups, columns, or other subsets of data.

- `dplyr::summarise()` – Calculate summary statistics
- `dplyr::group_by()` – Group observations
- `dplyr::across()` – Apply functions across multiple columns
- `aggregate()` – Grouped summary statistics
- `by()` – Apply functions to subsets
- `tapply()` – Apply functions by groups
- `vapply()` – Type-safe vectorized application
- `purrr::map()` – Apply functions across objects
- `data.table::.(...)` – Efficient grouped calculations with `data.table`

2.7 Additional Useful Packages Packages that provide additional functionality for descriptive statistics and summary analysis.

Package	**Main Purpose**
`psych`	Comprehensive descriptive statistics
`DescTools`	Statistical summaries and descriptive measures
`Hmisc`	Descriptive statistics and weighted summaries
`skimr`	Quick, readable dataset summaries
`summarytools`	Detailed descriptive reports
`janitor`	Frequency tables and data cleaning
`moments`	Skewness and kurtosis
`e1071`	Statistical moments and other statistical tools
`matrixStats`	Efficient row/column statistics
`dplyr`	Grouped and flexible summaries

2.8 Recommended R Functions Reference The following is a consolidated list of useful R functions for descriptive statistics.

```
# Central tendency
mean()
median()
weighted.mean()
DescTools::Mode()

# Dispersion
var()
sd()
IQR()
mad()
range()
min()
max()
DescTools::CoefVar()

# Position and distribution
quantile()
rank()
fivenum()
summary()
psych::describe()
skimr::skim()

# Distribution shape
moments::skewness()
moments::kurtosis()

# Frequency and categorical summaries
table()
prop.table()
xtabs()
addmargins()
janitor::tabyl()

# Grouped summaries
dplyr::group_by()
dplyr::summarise()
dplyr::across()
aggregate()
tapply()

# Weighted summaries
weighted.mean()
Hmisc::wtd.mean()
```

3. Distribution Analysis

4. Data Visualization and Exploratory Data Analysis

5. Missing Data Analysis

6. Outlier and Influential Observation Detection

7. Assessing Distributional Assumptions

8. Assessing Variance and Homogeneity Assumptions

9. Assessing Independence and Dependence

10. Assessing Relationships Between Variables

11. Data Transformations and Preprocessing

12. Sample Size and Data Adequacy

Comparing Groups (Means/Medians)

Relationships and Associations

Regression, Statistical Modeling and Prediction

Model Building, Diagnostics and Comparison

Visualization/ Presentation of Results

One-Sample t-Test

`Use when:`

Wilcoxon Signed-Rank Test

Predictive Modeling and Machine Learning

Time Series and Longitudinal Analysis

Survival and Event-Time Analysis

Multivariate and Dimension Reduction Techniques